WILDER GILMER ESPINOZA LUNA

Advanced Analytics | Credit Risk | Machine Learning Engineering

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LinkedIn GitHub HackerRank Portfolio GitLab

Professional profile

Statistics student at Universidad Nacional Mayor de San Marcos (UNMSM), focused on turning business problems into measurable machine learning solutions for banking: probability of default (PD) models, scorecards and credit approval policies. I prioritize simple, maintainable solutions, business metrics and rigorous validation. I have built MLOps workflows with CI/CD, model registries, champion/challenger and drift monitoring. In operations, I automated reports that reduced processing time by 83% and reduced recording errors by 28%. I use Python, SQL and PySpark, with hands-on Databricks experience and foundational knowledge of Google Cloud and Azure.

Technical skills

Banking and risk: PD, credit scoring, WoE/IV scorecards, cost-sensitive cut-offs, approval policies, loan loss provisioning, backtesting, KPIs and business metrics.
Modeling and statistics: regression, classification, forecasting, time series, feature engineering, out-of-time validation, data leakage prevention, ROC-AUC, KS, Gini, calibration, DeLong and Diebold-Mariano tests, Holm correction and SHAP.
Machine learning: scikit-learn, CatBoost, XGBoost, LightGBM, Optuna, PyTorch.
Data and cloud: Python, PySpark, SQL, R, pandas, ETL/ELT; Databricks (hands-on), Google Cloud Run, Azure (foundations) and AWS (basic knowledge).
ML engineering and AI: MLflow, champion/challenger, drift detection, retraining, CI/CD, FastAPI, Docker, GitHub Actions, LLMs, RAG, embeddings, agents, MCP, Power BI and Streamlit.

Professional experience

Warehouse Supervisor | Mebol 2024–2025 (6 months)

- Reduced inventory recording errors by 28% through validation rules implemented in SQL with the purchasing and sales teams.
- Automated weekly reports with SQL and Excel, cutting processing time from 6 hours to 1 (83%) and freeing 5 hours for stock analysis.
- Built Power BI dashboards with availability and traceability KPIs for more than 450 products.

Selected projects

Credit Risk Platform | Predictive modeling and MLOps Sep–Oct 2026

Python, CatBoost, XGBoost, LightGBM, Databricks, MLflow, FastAPI, SHAP

Repository Demo

- Modeled PD using 887,000 Lending Club loans, out-of-time validation and exclusion of 22 post-disbursement variables. Test ROC-AUC was 0.691 versus 0.681 for the WoE/IV benchmark; Gini 0.382 and KS 0.277.
- In the evaluation, a 0.18 threshold and 5:1 FN:FP cost ratio approved 61.3% of applications and reduced default in the approved portfolio from 15.5% to 9.6%. These are not live portfolio results.
- Implemented a Databricks MLOps workflow with CI/CD, model registry and champion/challenger. A temporal replay triggered two retraining runs; the challenger was not promoted without statistical evidence ($p = 0.61$). Includes 169 automated tests.

GeoBanca Norte Perú | Spatial analytics for financial services Oct 2026

GeoPandas, PySAL, spatial models, kriging, SHAP, integer optimization, QGIS

Repository Interactive poster

- Integrated financial, demographic and climate data across 327 northern Peruvian districts to assess financial access and climate risk.
- The spatial Durbin model achieved AIC 1648 versus 1696; kriging reduced cross-validation RMSE by 23% versus IDW. MCLP optimization proposed 25 service points covering 63,643 adults.

RiskRAG | Generative AI for risk and insurance regulations Oct 2026

RAG, BM25, FastAPI, PostgreSQL, pgvector, Terraform, Docker, MCP

Repository Demo

- Designed an SBS regulatory assistant with hybrid RAG, verifiable citations and abstention without evidence; classification and provisioning use deterministic rules.
- The system combines vector search and BM25, RRF fusion and reranking; it exposes regulations, financial calculations and the PD model as permission-separated MCP tools.
- Evaluated 89.6% citation precision, 91.7% intent detection and 100% adversarial containment on the evaluation set; 88 tests. Azure infrastructure was validated but not deployed.

Hackathons and competitions

Rumbo | AI-assisted job matching Jul–Aug 2026

All Things Agentic Hackathon, Google Cloud | Four-person team

Project Video

- Collaborated on a platform that normalizes vacancies, evaluates fit, identifies skill gaps and generates career paths.
- Co-designed three Gemini and Google ADK agents and a statistical matching score; a pipeline with embeddings and one batch LLM call was deployed on Cloud Run with FastAPI and React.

Orchestrate | OpenAI Œ HackerRank | Bronze medal Oct 2026

Global top 54 among over 60,000 participants | 72.1/100

Repository Event

- Built an agent in 24 hours to audit data-erasure requests and generate remediation plans.
- Designed an auditable architecture using Tesseract OCR as the authoritative source, Gemini for corroboration and SHA-256 caching for offline reproducibility.
- Achieved 32/32 correct public sample cases, step precision 1.000, recall 0.979 and F1 0.990; zero contract violations across 518 requests.

Education

Statistics studies | Universidad Nacional Mayor de San Marcos 2022–2027, in progress

PRONABEC scholarship recipient (2024) for academic performance.

GCI World 2026 | Matsuo-Iwasawa Laboratory, The University of Tokyo Sep 2026–present

International data science and machine learning program.

Certifications and languages

Certifications: Advanced ML Operations (Databricks); MLOps Specialization (Duke University); Machine Learning Specialization (DeepLearning.AI and Stanford).

Spanish: native **English:** Advanced English as a Second Language, Saylor University.